

How the index is actually computed

SCITI is a two-layer model. First, structured research assigns seven pillar scores from auditable outcome indicators, evidence items, and field verification. Second, deterministic mathematics aggregates those pillar scores into a composite score, tier, and contribution breakdown.

METHOD CODE SCITI-2026-METH · RELEASE APRIL 2026 · DATA CUT-OFF 2026-04-04 · AUDITED LOCAL REPO STATE

SCORE DOMAIN

0-100

All seven pillar inputs are validated against the same closed numeric interval before aggregation.

TIER THRESHOLDS

Alpha ≥ 65 · Beta ≥ 45 · Gamma < 45

Once the composite score is fixed, tier assignment is automatic. No hidden committee override exists in the scoring layer.

EVIDENCE REGISTRY

14 source families

The provenance registry tracks government, satellite, international reference, and field-verification sources.

PLATFORM MAP

12 public endpoints

A separate registry maps public data platforms and dashboards that researchers and cities can inspect directly.

1. Analytical Structure

The critical methodological distinction is between **research judgment** and **deterministic aggregation**.

- Research layer: each city receives seven pillar scores from a structured reading of public indicators, evidence items, and field verification.
- Aggregation layer: once those seven inputs are fixed, the composite score, weighted contribution bars, and tier assignment are fully reproducible from code.
- Confidence layer: evidence quality is reported separately so uncertainty stays visible instead of being hidden inside the score.

SCITI therefore does not claim that one API feed or one normalization function can fully explain urban life. It claims something narrower and stronger: the final aggregation step is exact, while the research step is explicit about where judgment remains necessary.

2. Composite Mathematics

The composite score is a weighted average over seven pillars.

Composite = $\Sigma(\text{pillar score} \times \text{weight}) / 100$

PILLAR	WEIGHT	INTERPRETIVE ROLE
Livability	25%	Daily urban function: housing, mobility, utilities, and whether life works without friction.
Economy	20%	Productive capacity, income, output, and honest demand absorption.
Safety	15%	Crime exposure, emergency readiness, and practical security.
Wellbeing	15%	Depth of public-service capacity after the ribbon-cutting.
Environment	10%	Air, water, green cover, and ecological resilience.
Hospitality	10%	Belonging, civic warmth, cultural life, and whether people want to stay.
Digital	5%	Enabling infrastructure: data rails, IoT, and digital service use that changes outcomes.

The weights sum to 100. Livability carries the largest share because a city must function for residents before it can meaningfully claim to be smart. Digital carries the smallest share because technology is an enabler, not the goal itself.

3. Data Confidence Mathematics

Confidence is reported separately from the composite score. It is designed to answer a different question: *how dense and fresh is the evidentiary backbone supporting this dossier?*

Confidence = weighted average(core coverage 65%, extended coverage 15%, provenance density 10%, freshness 10%)

- **Core coverage (65%):** six core indicators are tracked across the repo-level city metric model — GPP per capita, average monthly income, annual PM2.5, hospital beds per 10K, crime per 100K, and green coverage.
- **Extended coverage (15%):** seven additional indicators widen the baseline — GPP growth, PM2.5 trend, water quality, forest coverage, FDI inflow, industry composition, and labor force.
- **Provenance density (10%):** the model rewards cities whose score narrative is backed by a thicker stack of observations and source-linked supports.
- **Freshness (10%):** observations are weighted against the fixed release cut-off of 2026-04-04 rather than against a moving runtime date, keeping the release reproducible.

The confidence label is then classified as High, Medium, or Low. Crucially, this label does not secretly alter the composite score. It stays visible as a separate signal.

4. Evidence Scope

The methodology relies on a registry of evidence source families and a separate platform map. The current audited release recognizes the following implemented source families and public endpoints in the repository.

Core evidence families used or referenced by the provenance layer

- National Statistical Office (NSO)
- National Economic and Social Development Council (NESDC)
- Pollution Control Department / Air4Thai
- Royal Thai Police
- Ministry of Public Health
- GISTDA satellite / geospatial data
- Open-Meteo / Copernicus cross-checks
- depa smart city registry
- citydata.in.th platform references
- Board of Investment (BOI)
- ONEP environmental performance inputs
- Royal Forest Department
- International contextual reference sources
- Field verification

Mapped public data platforms and endpoints

- data.go.th
- citydata.in.th
- Air4Thai
- DOPA population statistics
- NSO statistical hub
- GISTDA Sphere
- TMD weather platform
- NESDC economic releases
- BOI IPStat
- ONEP environmental performance releases
- Royal Forest Department datasets
- HDX administrative boundaries

Not every source contributes to every city or every pillar. The method is multi-source by design, but it does not fabricate a false sense of uniform observability across all cities.

5. Treatment of Thinly Observed Cities

Promotion-zone and registered cities with insufficient evidence are not dressed up as fully observed cases. In the repository they receive provisional low-confidence baselines and are clearly marked as such. This preserves rank continuity while making their uncertainty explicit.

That is an uncomfortable but methodologically cleaner choice than pretending the missing evidence does not matter.

6. What the Method Does Not Claim

- No hidden caps, prestige bonuses, or bespoke city penalties in the aggregation step.
- No claim that a single normalization pipeline fully explains city quality.
- No claim that every listed source is used equally for every city.
- No promise that scores say anything about events after the release cut-off.

Those limits matter. A methodology becomes more credible, not less, when it is precise about what is automated, what is curated, and what remains uncertain.

7. Standards Alignment

- UN-Habitat City Prosperity Initiative (CPI)
- ISO 37122:2019 — Indicators for Smart Cities
- UN SDG 11 — Sustainable Cities and Communities
- ASEAN Smart Cities Framework (ASCF)
- ASEAN Smart Cities Action Plan (ASCAP) 2021-2025
- New Urban Agenda (2016)
- Thailand PPP Act B.E. 2562 (2019)

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